

Eliminating On-Fault Normal-Stress Speckle in Pure-Upwind DG Dynamic Rupture

Why over-integration and slip-rate resampling (the SeisSol recipe) remove the speckle —
mechanism, equations, and an implementation plan for the matrix/pure-upwind path
(TPV31)

SEAS-MFEM dynamic-rupture team

2026-06-02

Contents

Purpose and scope	2
1. The symptom, and what we have already ruled out	2
2. The one equation both codes share, and why σ_n must be constant	3
3. Root cause: aliasing of the nonlinear friction	4
3.1 The friction law is the only nonlinearity, and it is not a polynomial	4
3.2 What “aliasing” means here, precisely	4
3.3 Two channels: per-step flux aliasing, and secular state accumulation	5
3.4 The chain from friction aliasing to the σ_n speckle and drift	5
3.5 Why raising the order does not help (and can hurt)	6
4. SeisSol’s two mechanisms (with equations)	6
4.1 Over-integration of the fault flux	6
4.2 Resampling: keep the accumulated slip/state in the degree- N space	8
4.3 A worked 1-D illustration (what each technique actually does)	9
4.4 Where SeisSol applies friction in time (context, not the lever)	9
5. Our current code vs. the target (the exact gap)	10
6. Implementation plan (for /code-implement after review)	10
Phase 0 — Local reproduction harness (no production runs)	10
Phase 1 — Over-integration of the fault flux (the prerequisite)	11
Phase 2 — Resample operator R on the fault face	11
Phase 3 — Apply R in the friction iterators	11
Phase 4 — Wire into the spatial driver + matrix path	12
Acceptance criteria	12

7. Validation plan	12
8. Risks, uncertainties, and fallback	13
9. References	13

Purpose and scope

This document explains, from first principles and with equations, **why** the two SeisSol techniques — **spatial over-integration of the fault flux** and **slip-rate / state-variable resampling** — eliminate the on-fault normal-stress “speckle” (and the secular σ_n drift it drives) that our pure-upwind discontinuous-Galerkin (DG) dynamic-rupture runs exhibit.

It is written to answer one question that has not yet been made concrete:

*We can see that mixed flux removes the speckle and that SeisSol stays clean with pure upwind, but **how** do over-integration and resampling actually eliminate the speckle values we observe?*

The target application is **TPV31**, which is forced onto **pure upwind + ADER + matrix (heterogeneous-material) Riemann** — the matrix Riemann solver is needed for TPV31’s depth-varying material, *not* a fault-normal material contrast (the material is identical on both sides at each depth) — and can use *neither* mixed flux *nor* the RK integrator. The two techniques here are **flux-agnostic and ADER-compatible**: they change *how the nonlinear friction is represented*, not the flux or the eigenvalues, so they apply unchanged to that path. The same fix benefits every pure-upwind spatial-driver run (TPV102/104/205, SAFS).

This is an explanation + implementation plan for review. **No source code is changed by this document.**

1. The symptom, and what we have already ruled out

On a planar fault with material identical on both sides (TPV31, TPV102/104/205), the on-fault normal stress σ_n is **analytically constant** for a strike-slip rupture. In our pure-upwind runs it instead shows:

- a high-spatial-frequency **speckle** across the fault plane (the ParaView fault-surface field, amplitude comparable to σ_{n0}), and
- a secular σ_n **collapse** at every station (TPV102: $120 \rightarrow 80, -30, 75$ MPa; TPV31: $60.6 \rightarrow 23$ MPa, then blow-up).

Established by controlled experiment and code reading (see the companion findings in `debug_document/tpv31_debug_document/`):

observation	conclusion
homogeneous TPV102/104/205 (scalar Riemann) drift identically	not the heterogeneous-material / matrix path
mixed flux (touches only fault- <i>adjacent bulk</i> faces) removes it	the fault-face friction formula is fine; the seed is near-fault
serial $\equiv$ parallel to 10^{-14} (<code>test_rupture_tilted_fault_serial_vs_parallel</code> , $\theta = 0$)	intrinsic , present in serial; not MPI / partition
planar fault $\Rightarrow n \cdot \hat{n}_{\text{ref}} = 1$, frame exact	not the orientation/frame bug (that is curvilinear-only)
p1 $\approx$ p2 (erratic, non-monotone); clean mixed-flux p1 $\approx$ p2	not “just raise the order” — order does not converge it

So the defect is **intrinsic to the discretization, present at p1 in serial, in the common (scalar + matrix) pure-upwind path, and order-independent**. This document identifies that defect as **aliasing of the nonlinear friction law**, and explains why SeisSol’s over-integration + resample remove it.

2. The one equation both codes share, and why σ_n must be constant

At each fault evaluation point, in fault-aligned coordinates (n, t_1, t_2) , both our code (`fault_face_flux.cpp:62`) and SeisSol (`FrictionSolverCommon.h:182`) compute the identical Godunov interface state:

$$\sigma_n^* = \eta_p \left(\underbrace{v_n^- - v_n^+}_{\llbracket v_n \rrbracket} + \frac{\sigma_n^+}{Z_p^+} + \frac{\sigma_n^-}{Z_p^-} \right), \quad \eta_p = \frac{Z_p^+ Z_p^-}{Z_p^+ + Z_p^-}, \quad Z_p = \rho c_p.$$

$$\tau_i^* = \eta_s \left(\llbracket v_{t_i} \rrbracket + \frac{\tau_i^+}{Z_s^+} + \frac{\tau_i^-}{Z_s^-} \right), \quad i = 1, 2.$$

The friction law then solves for a slip rate S from the shear balance and returns the **corrected (radiated) shear traction**

$$\tau_i^{\text{corr}} = \tau_i^* - \eta_s S_i,$$

leaving the normal traction at its Godunov value. Writing $\sigma_n^{\text{tot}} = \sigma_{n0} + \sigma_n^*$ (perturbation about the background), the normal-stress perturbation is

$$\delta\sigma_n = \eta_p \llbracket v_n \rrbracket + \eta_p \left(\frac{\sigma_n^+}{Z_p^+} + \frac{\sigma_n^-}{Z_p^-} \right)$$

For a planar strike-slip fault with symmetric material there is **no normal-direction wave**: the physical $\llbracket v_n \rrbracket = 0$ and the normal-stress fluctuation is zero, so $\delta\sigma_n \equiv 0$. **Any** spurious normal-velocity jump is amplified by $\eta_p \approx Z_p/2 \approx 8 \times 10^6 \text{ Pa s/m}$:

$$\delta\sigma_n \approx \eta_p \llbracket v_n \rrbracket \implies \llbracket v_n \rrbracket \approx \frac{40 \text{ MPa}}{8 \times 10^6} \approx 5 \text{ m/s}.$$

The observed $\approx 40 \text{ MPa}$ drift therefore corresponds to a **coherent** $\llbracket v_n \rrbracket \approx 5 \text{ m/s}$ — *of the same order as the slip rate itself*. This is not faint round-off grit; it is a large, systematic leak of shear motion into the normal channel. The rest of this document explains where that coherent $\llbracket v_n \rrbracket$ comes from.

3. Root cause: aliasing of the nonlinear friction

3.1 The friction law is the only nonlinearity, and it is not a polynomial

In a DG scheme the bulk operator for linear elasticity is **linear**: on affine elements with exact quadrature, the nodal and modal formulations are *identical* (a change of basis). The **only** nonlinearity in the entire dynamic-rupture operator is the fault friction. At a fault point it returns

$$S = G(\tau^*, \sigma_n^*, \psi, \dots), \quad \tau^{\text{corr}} = \tau^* - \eta_s S,$$

where G is rate-and-state (an asinh form), or linear slip-weakening with a **kink** at $\delta = d_c$, or a rupture front where S jumps from 0 to several m/s over one element. **None of these is a polynomial**. Even when the inputs $\tau^*(x), \sigma_n^*(x), \psi(x)$ are degree- N polynomials on the face, the output $S(x)$ has spectral content at **all** wavenumbers, including far above degree N .

3.2 What “aliasing” means here, precisely

A DG scheme can only carry the degree- N projection of S . The L^2 projection onto the degree- N face polynomial space is

$$\widehat{S} = \Pi_N S = M^{-1} \int_F \phi S(x) dx, \quad M_{ij} = \int_F \phi_i \phi_j dx,$$

with $\phi = \{\phi_i\}$ the basis. The integral is evaluated by **quadrature**,

$$\int_F \phi_i S dx \approx \sum_q w_q \phi_i(x_q) S(x_q).$$

Because S is **not** a polynomial, *no finite quadrature is exact*. A rule that is exact only to degree $2N$ (our minimal “mass-matrix order”, $2 \times \text{order}$) cannot distinguish the high-wavenumber content of S (modes above N) from the resolved low modes: that energy is **folded down** — **aliased** — **onto the coefficients** $\widehat{S}_0, \dots, \widehat{S}_N$. This is aliasing error. It is *not* reduced by mesh refinement of a fixed feature and it is *not* zero-mean in the quantities that matter (Section 3.4).

3.3 Two channels: per-step flux aliasing, and secular state accumulation

The friction nonlinearity pollutes the solution through **two distinct channels**, and the two SeisSol techniques each close one of them.

(1) Per-step flux aliasing (instantaneous). Each step the radiated traction is assembled as the flux integral $\int_F \phi_i \tau^{\text{corr}} dx$, evaluated by quadrature. Because τ^{corr} is non-polynomial (Section 3.1), an under-integrated rule **folds its $> N$ content onto the resolved coefficients $0..N$** (Section 3.2). The fold-over hits the **top coefficient c_N first** — it has the lowest fold-over threshold (Section 4.1) — so the spurious energy appears as the most oscillatory representable mode, degree N : *the per-element checkerboard you see as speckle*. **Over-integration closes this channel:** with enough Gauss points the resolved coefficients are the true L^2 values and no spurious top-mode energy is injected. This keeps the solution's *genuine* degree- N content — it removes the *spurious* part, it does **not** delete the mode.

(2) Secular state accumulation. Slip and state are advanced every step by integrating the *pointwise* friction output at the GPs. Those pointwise values carry $> N$ content that the degree- N DOF field cannot represent; left in, it **accumulates in the slip/state field over many steps**, drifting it out of the polynomial space and re-seeding speckle through the strength $G(\cdot)$. **Resampling closes this channel:** it projects the per-step slip/state increment back onto the degree- N DOF space (Section 4.2), discarding only the unrepresentable $> N$ part and keeping all modes $0..N$ (including mode N).

The two are **coupled, not independent**: the resample projector is the *identity* whenever the GP count equals the DOF count (the minimal mass-matrix rule), so it does nothing **until** the flux is over-integrated. Over-integration is the prerequisite; resample is the secular cleanup layered on top.

3.4 The chain from friction aliasing to the σ_n speckle and drift

Here is the causal chain the user asked to see spelled out:

$$\underbrace{S(x), \tau^{\text{corr}}(x)}_{\text{nonlinear friction}} \xrightarrow[\text{no dealiasing}]{\Pi_N} \underbrace{\text{spurious degree-}N \text{ top mode in } \tau^{\text{corr}}}_{\text{checkerboard on the fault}} \xrightarrow[\text{radiates}]{\text{bulk}} \underbrace{\text{checkerboard wavefield, incl. } v_n}_{\text{spurious normal P-radiation}} \xrightarrow{\eta_p}$$

1. The friction reaction $\tau^{\text{corr}} = \tau^* - \eta_s S$ is the traction the fault **applies back to the bulk**. If τ^{corr} carries the spurious top mode, the fault **radiates a checkerboard wavefield**.
2. A checkerboard shear source on an imperfect (non-mirror-symmetric) near-fault mesh does **not** radiate purely tangentially: it leaks a fault-normal particle-velocity component v_n into the bulk next to the fault.
3. That v_n returns to the fault as $\llbracket v_n \rrbracket \neq 0$ and is multiplied by $\eta_p \approx 8 \times 10^6$ in Eq. (box, Section 2) — turning a small slip-rate ringing into a **tens-of-MPa σ_n speckle**.

Why a zero-mean speckle becomes a one-way drift. The friction *strength* uses $|\sigma_n|$ (rate-state) or $\max(\sigma_n, 0)$ (LSW). These are **even / rectifying** functions. A symmetric $\pm \delta \sigma_n$ speckle therefore produces an **asymmetric** change in strength, hence in slip, hence a **net (secular) bias** in σ_n — the observed collapse. For LSW, once a tensile excursion crosses $\sigma_n \leq 0$ the strength drops to cohesion,

the fault free-slides, and the bias runs away (TPV31 blow-up). **Remove the speckle and both the speckle and its rectified drift disappear.**

3.5 Why raising the order does not help (and can hurt)

Raising N makes the rupture front *sharper relative to the element* and pushes the aliased energy onto a higher-wavenumber top coefficient c_N . With no degree-dependent dealiasing, more order folds *more* $> N$ energy onto the resolved modes — exactly the erratic, non-converging p1→p2 behaviour we measured. SeisSol’s cure is **order-aware**: the fault over-integration scales with the order $((N + 2)^2$ GPs) and the resample projects onto the *current* degree- N space, so aliasing is controlled at every order and high-order runs pay off; ours has neither knob, so nothing tracks the order.

4. SeisSol’s two mechanisms (with equations)

SeisSol uses pure-upwind Godunov flux **everywhere** (no mixed flux) and stays clean because of *how it forms and represents the fault solution*:

4.1 Over-integration of the fault flux

SeisSol integrates the fault flux at $(N + 2)^2$ Gauss points per triangular face (Pelties et al. 2014, §3 — “these flux functions are integrated with a quadrature based on $(N + 2)^2$ GPs”), and assigns friction parameters per Gauss point. This is **more** than the minimal mass-matrix rule. The extra points capture more of the spectrum of the nonlinear integrand, so the aliasing of high modes onto the resolved coefficients (Section 3.2) is strongly suppressed:

$$\left| \sum_q w_q \phi_i(x_q) S(x_q) - \int_F \phi_i S dx \right| \xrightarrow{\text{more GPs}} \text{small.}$$

Our path uses `IntRules.Get(geom, 2*order)` (`spatial_dyn_driver.cpp:217`) — the minimal mass-matrix order, **under-integrated for the nonlinear friction**.

Worked example: how under-integration makes a high mode masquerade as a resolved one

This is the single most important fact, so here it is in numbers. Work on the reference interval $[-1, 1]$ with the Legendre basis of a degree- $N = 2$ element,

$$P_0 = 1, P_1 = x, P_2 = \frac{3x^2-1}{2}, P_3 = \frac{5x^3-3x}{2}, P_4 = \frac{35x^4-30x^2+3}{8},$$

whose defining property is orthogonality — *different modes integrate to zero against one another*:

$$\int_{-1}^1 P_m(x) P_n(x) dx = 0, \quad m \neq n.$$

A degree-2 element is therefore **blind** to any higher mode: the projection of $P_3, P_4, \dots$ onto $\{P_0, P_1, P_2\}$ is exactly zero.

What our actual rule is. Our fault quadrature is `IntRules.Get(geom, 2*order)`, a rule exact to degree $2N$. For $N = 2$ MFEM returns the **3-point Gauss rule** (nodes $0, \pm\sqrt{3/5}$, weights $\frac{8}{9}, \frac{5}{9}, \frac{5}{9}$; exact to degree 5). The contaminated coefficient that matters is the **top** one, $c_2 = \int_{-1}^1 P_2 S dx$ — the fold-over reaches it first.

The next mode up, P_3 , does NOT alias under our rule. With $S = P_3$ the product $P_2 P_3$ has degree 5, which the 3-point rule integrates *exactly*, so

$$c_2^{(3\text{-pt})} = \int_{-1}^1 P_2 P_3 dx = 0 \quad \checkmark$$

— the true value, by orthogonality. (So, unlike a naive degree- $2N-1$ rule, our actual rule is already clean for P_3 ; the first mode that *bites* is one higher.)

Put the first biting mode into the slip rate: $S(x) = P_4(x)$. The true top coefficient is zero ($P_4 \perp P_2$): $c_2 = \int_{-1}^1 P_2 P_4 dx = 0$. **Now compute it with our 3-point rule.** The integrand $P_2 P_4$ has degree $6 > 5$, so the rule is *not* exact. At $x = \pm\sqrt{3/5}$ ($x^2 = \frac{3}{5}$): $P_2 = \frac{2}{5}$, $P_4 = -\frac{3}{10}$, so $P_2 P_4 = -\frac{3}{25}$; at $x = 0$: $P_2 = -\frac{1}{2}$, $P_4 = \frac{3}{8}$, so $P_2 P_4 = -\frac{3}{16}$. Hence

$$c_2^{(3\text{-pt})} = 2 \cdot \frac{5}{9} \left(-\frac{3}{25} \right) + \frac{8}{9} \left(-\frac{3}{16} \right) = -\frac{2}{15} - \frac{1}{6} = \boxed{-\frac{3}{10}} \neq 0.$$

The unresolved P_4 mode has **injected** $-\frac{3}{10}$ **into the resolved P_2 slot**. The 3-point rule cannot tell P_4 from P_2 at its three nodes and charges P_4 's integral to P_2 . *That nonzero c_2 is the spurious top-mode wiggle — the speckle.*

Over-integrate with a 4-point Gauss rule (exact to degree $7 \geq 6$): the degree-6 product is now integrated exactly, so

$$c_2^{(4\text{-pt})} = 0 \quad \checkmark$$

and P_4 again contributes nothing to the resolved modes. Over-integration did **not smooth** P_4 (P_4 is untouched) — it stopped P_4 from *masquerading* as P_2 . In one line: $-\frac{3}{10}$ (contaminated) $\rightarrow 0$ (clean).

The general rule. A quadrature exact to degree D evaluates $\int_{-1}^1 P_k S dx$ correctly *only* for the part of S of degree $\leq D - k$; anything above that **folds onto** the coefficient c_k . The $(N+1)$ -point Gauss rule that `IntRules.Get(geom, 2*order)` returns is exact to degree $D = 2N + 1$ (Gauss over-delivers by one beyond the requested $2N$), so c_N is correct up to degree $D - N = N + 1$ and the first mode that folds onto it is $\deg(P_N P_{N+2}) = 2N + 2 > 2N + 1$, i.e. P_{N+2} (for $N = 2$: P_4 , as above). Over-integration raises D and pushes the fold-over threshold above the modes that bite. (*On simplex faces MFEM may deliver exactly degree $2N$ rather than $2N + 1$, in which case the first folding mode is P_{N+1} ; tie the bound to the rule actually returned for the face geometry in use.*)

Why it is called “folding” (the sampling view). On M equally-spaced points $x_j = 2\pi j/M$,

$$e^{i(k+M)x_j} = e^{ikx_j} \underbrace{e^{i2\pi j}}_{=1} = e^{ikx_j},$$

so mode $k + M$ is *identical to mode k at the samples*, and the measured coefficient is the true one plus all of its aliases:

$$\widehat{S}_k^{(\text{sampled})} = \widehat{S}_k + \widehat{S}_{k+M} + \widehat{S}_{k-M} + \dots$$

More points $M \Rightarrow$ the aliases $k \pm M$ sit farther away $\Rightarrow$ the resolved band stays clean. (“A 1 kHz tone read as 100 Hz” is exactly $e^{i(k+M)x_j} = e^{ikx_j}$.)

This $c_2 = -\frac{3}{10}$ is precisely the spurious top-mode oscillation in the flux that, via $\tau^{\text{corr}} = \tau^* - \eta_s S$, radiates the checkerboard and feeds $\delta\sigma_n = \eta_p (v_n^- - v_n^+)$. **Over-integration removes this contamination of the resolved modes — it is the per-step lever on the speckle.** Resampling (next) plays the *complementary, secular* role: it projects the *accumulated* slip/state increment back onto the degree- N space so the unrepresentable $> N$ content does not build up over many steps. Resampling **keeps** the top mode c_N (the genuine degree- N content); it is the over-integration, not the resample, that makes the per-step traction clean.

4.2 Resampling: keep the accumulated slip/state in the degree- N space

After the pointwise friction solve, SeisSol **resamples** the per-step increment of the accumulated state — the **slip-rate magnitude** for LSW (`LinearSlipWeakening.cpp:18`), the **state-variable increment** $\Delta\psi$ for rate-state (`FastVelocityWeakeningLaw.h:144`, `RateAndState.h:89`) — onto the DOF polynomial space and re-evaluates at the (over-integrated) fault GPs (matrix doc `LinearSlipWeakening.h:227`):

“resampleMatrix first projects LocSR on the two-dimensional basis on the reference triangle with degree $\leq$ ConvergenceOrder -1 , and then evaluates the polynomial at the quadrature points.”

This is the full degree- N space, not $N - 1$. In SeisSol the polynomial degree equals ConvergenceOrder -1 (`FreeSurfaceIntegrator.h:42`: `PolyDegree = ConvergenceOrder - 1`), so “degree $\leq$ ConvergenceOrder -1 ” means “degree $\leq N$ ” — the *same* space as the DOFs. The comment states the goal exactly: that “the state increment (slip) lies in the **same polynomial space as the degrees of freedom**.” Numerically, the generated resample matrices have rank $(N + 1)(N + 2)/2$ — the full degree- N DOF count — at every order (verified for $N = 1..6$).

Formally, with V_N the Vandermonde from degree- N modal coefficients to the (over-integrated) GP values and $W = \text{diag}(w_q)$ the quadrature weights,

$$\boxed{R = V_N (V_N^\top W V_N)^{-1} V_N^\top W} \quad \Delta \leftarrow R \Delta,$$

applied to the increment Δ (slip-rate magnitude, or $\Delta\psi$). R is the L^2 **projector onto the degree- N DOF space, expressed in GP-value space**: it **reproduces all modes $0..N$ exactly (the top mode N included)** and **discards only the $> N$ content** that the over-integrated GPs expose but the DOF basis cannot represent. Two consequences:

- **R is the identity when $\#\text{GP} = \#\text{DOF}$** (the minimal mass-matrix rule, e.g. tensor Gauss on a quad face: $(N + 1)^2$ GPs = $(N + 1)^2$ DOFs). It does nothing until the flux is over-integrated (Section 4.1) — the two techniques are **coupled**.
- **R does not touch the radiated traction.** SeisSol builds $\tau^{\text{corr}} = \tau^* - \eta_s S$ (and the directional

slip) from the *un-resampled* friction-solve rate inside `calcSlipRateAndTraction`, *before* the resample; resample is applied only to the accumulated slip/state that drives the strength on the next step. So R controls the **secular** out-of-space drift of the state, while **over-integration** controls the **per-step** radiated speckle (Sections 4.1 and 3.3).

A caveat on naming. SeisSol’s `BiMaterialFault` specialization (Prakash–Clifton) is a true **across-fault material contrast**, and there SeisSol **disables** resampling because in that regime “resampling introduces artificial oscillations” (`LinearSlipWeakening.h`:261). **TPV31 is not this case** — its material is identical on both sides at each depth (our “matrix” Riemann handles depth-heterogeneity, not a fault-normal jump), so resampling is appropriate and should be **on** for TPV31/TPV102/104/205. The action item is the converse: on any target with a genuine across-fault material contrast (some SAFS configurations), resample must be **disabled**, matching SeisSol.

4.3 A worked 1-D illustration (what each technique actually does)

Take a single fault element as $[-1, 1]$, degree $N = 2$, Legendre basis P_0, P_1, P_2 . Let the friction response be a sharp rupture front, $S(x) = \frac{1}{2}(1 + \tanh(x/\varepsilon))$ with $\varepsilon = 0.1$. As a non-polynomial it has content at **all** degrees: $S = a_0P_0 + a_1P_1 + a_2P_2 + (\text{degree} > 2 \text{ tail})$.

- **Under-integrated flux (minimal GPs):** the > 2 tail folds onto the resolved coefficients (Section 4.1), worst onto the top coefficient a_2 , so the per-step flux integral $\int_F \phi_i \tau^{\text{corr}}$ is **wrong** — carrying a spurious, maximally-oscillatory degree-2 component. *Across many elements these spurious top modes tile into the checkerboard speckle.*
- **Over-integration (the per-step fix):** with enough GPs the resolved a_0, a_1, a_2 are the **true** L^2 coefficients — no folded-in energy. The radiated traction is the honest degree-2 projection of the friction, so it no longer carries a *spurious* top mode. (Its genuine a_2 — the true degree-2 content — is **kept**; we remove the aliasing, not the mode.)
- **Resample R (the secular fix):** the slip/state is advanced by integrating the *pointwise* friction at the over-integrated GPs, which still carries the > 2 tail. R projects that increment back onto degree 2 ($R\hat{\Delta} = a_0P_0 + a_1P_1 + a_2P_2$, top mode **kept**), discarding only the unrepresentable > 2 tail. Without it the tail accumulates in the slip/state over many steps and re-seeds speckle through the strength $G(\cdot)$; with it the state stays a clean degree-2 field.

Over-integration makes the per-step radiated traction the true degree-2 projection (no spurious top mode) $\Rightarrow$ no checkerboard radiation $\Rightarrow \llbracket v_n \rrbracket \rightarrow \text{round-off} \Rightarrow \delta\sigma_n = \eta_p \llbracket v_n \rrbracket \rightarrow 0$; resample keeps the *accumulated* state in the degree-2 space so the drift does not creep back. **That is how over-integration (per step) + resample (secular) eliminate the speckle — and why resample is a no-op without the over-integration underneath it.**

4.4 Where SeisSol applies friction in time (context, not the lever)

SeisSol solves friction at the N Gauss–Legendre *time* nodes of one ADER step and integrates with `timeWeights` to a single imposed state (`FrictionSolverCommon.h`:329; Uphoff thesis Eq. 4.60), with one slip rate per (space-GP, time-node) used for *both* the flux and the slip increment. Our isolation run (pure-upwind + RK45, job 7762948) showed the **time integrator is not the lever** — RK45 drifts the same as ADER. The spatial dealiasing (4.1, 4.2) is the lever.

5. Our current code vs. the target (the exact gap)

ingredient	SeisSol	SEAS-MFEM (pure-upwind ADER, spatial driver)
fault-flux quadrature	$(N + 2)^2$ GPs (over-integrated)	2*order (mass-matrix minimal) — spatial_dyn_driver.cpp:217
slip / state dealiasing	R : project the per-step increment to degree N (full DOF space) on the over-integrated GPs	none (grep resample = 0 hits)
resample target	LSW: slip-rate magnitude $\rightarrow$ accumulated slip; RS: state-variable increment $\Delta\psi$	none
fault state storage	per-GP, kept in a degree- N field via R	per-GP, independent (DOFData, no projection)

The two missing ingredients are exactly the dealiasing pair. Crucially, **both are flux-agnostic and do not alter the eigenvalues**, so they are compatible with ADER (no RK needed) and with the heterogeneous-material **matrix** Riemann path — i.e. with TPV31’s hard constraints. (Note this “matrix” path is for depth-heterogeneity, *not* a fault-normal material contrast; see the BiMaterialFault caveat in Section 4.2.)

6. Implementation plan (for /code-implement after review)

Goal: add (A) over-integration of the fault flux and (B) slip/state resampling to the pure-upwind fault path — **in that order** (resample is a no-op without over-integration, Section 4.2) — keeping byte-exact behaviour when both knobs are off, and validate that they flatten σ_n without mixed flux or RK.

Phase 0 – Local reproduction harness (no production runs)

- Build a **serial**, homogeneous, **planar-fault** mini-rupture unit test large enough for a front to propagate (e.g. a structured slab of ≈ 20 – 40 fault faces; the existing 2-tet fixture is too small to develop the drift).
- Outputs: $\max_F |\sigma_n - \sigma_{n0}|(t)$ and $\max_F \|\llbracket v_n \rrbracket\|(t)$.
- Acceptance: reproduces a growing σ_n excursion at p1 in serial (confirms Sections 1–3 locally and gives an A/B testbed). Run a deliberately **asymmetric** vs **mirror-symmetric** mesh variant to confirm the near-fault mesh-asymmetry sensitivity.

Phase 1 – Over-integration of the fault flux (the prerequisite)

- Add a separate, higher fault-quadrature degree (target $(N + 2)^2$ GPs per face, SeisSol’s choice, or a tunable `fault_overint_factor`) used for the friction solve **and** the flux-assembly integral $\int_F \phi_i F_h$, decoupled from the bulk mass-matrix rule.
- Gate behind `--fault-overint <k>` (default off $\Rightarrow$ falls back to $2 \times \text{order} \Rightarrow$ byte-exact with today).
- Assign friction parameters per over-integration GP. This is the **per-step speckle lever** (Section 4.1), and it must land **before** resample, because the resample projector is the identity unless $\#GP > \#DOF$.

Phase 2 – Resample operator R on the fault face

- For the reference fault-face geometry, order N , and the **over-integrated** GP set from Phase 1, build R (Eq. 4.2) once: GP-value $\rightarrow$ **degree- N L^2** projection $\rightarrow$ GP-value. Affine faces share one reference R (the $|J_F|$ scale cancels in the projector).
- New helper, e.g. `dynamic/fault_resample.hpp` (mirrors SeisSol’s `init::resample`), with a unit test asserting:
 - (i) R reproduces **any degree- N field exactly** — including the **pure degree- N top mode** (NOT annihilates it);
 - (ii) R **removes degree- $> N$ content** sampled at the over-integration GPs (the part orthogonal to the degree- N space goes to zero; the degree- N best fit remains);
 - (iii) R is idempotent ($R^2 = R$) with **rank** $(N + 1)(N + 2)/2$ (full degree- N DOF count — matches SeisSol’s resample matrices, verified $N = 1..6$);
 - (iv) at the minimal rule ($\#GP = \#DOF$), $R = I$ (so Phase 2 alone is byte-exact).

Phase 3 – Apply R in the friction iterators

- In `friction_substep_iterator.{hpp,cpp}` (and the RK path `rk_time_stepper`): build τ^{corr} and the directional slip from the **un-resampled** friction-solve rate (as today), then apply R **only to the accumulated state increment**, per face — matching SeisSol (`calcSlipRateAndTraction` runs before the resample). **Do not rebuild τ^{corr} from a resampled quantity.**
- **Resample target depends on the law** (Section 4.2): **LSW** $\rightarrow$ resample the slip-rate magnitude and integrate it into the accumulated slip (which drives the slip-weakening μ); **rate-state** $\rightarrow$ resample the state-variable increment $\Delta\psi$.
- This needs a **per-face gather/scatter** of the per-QP `DOFData`; the QPs of one face form the polynomial R acts on. The data are already laid out in per-face contiguous QP blocks of size `nbF = IntRules.Get(geom, 2*order).GetNPoints()` (`spatial_dyn_driver.cpp:217`); over-integration grows `nbF`. Assert QP-count-per-face = rule size.
- Gate behind `--fault-resample` (default off). With Phase 1 also off, $R = I$ and the path is byte-exact with today.

```
// Phase 3 sketch. tau_corr + directional slip come from the (un-resampled)
// friction-solve rate, exactly as today. Resample acts ONLY on the accumulated
// state increment, and only does anything when the GP set is over-integrated
// (Phase 1); off => byte-exact. DOFData::V1/V2 = friction-solve slip rate; the
// *_trial values live in the per-QP working state (fault_face_flux.hpp:153).
```

```

for (const Face &f : fault_faces) {
    for (int q = 0; q < nbf; ++q) { // nbf = QPs per face (grows w/ over-int)
        DOFData &d = dof[f.qp0 + q];
        d.slip1 += d.V1 * dt_sub; // un-resampled rate -> directional slip
        d.slip2 += d.V2 * dt_sub;
        d.tau1_corr = work[q].tau1_trial - d.eta_s * d.V1; // un-resampled rate -> flux
        d.tau2_corr = work[q].tau2_trial - d.eta_s * d.V2;
    }
    if (fault_resample) { // dealias the ACCUMULATED state increment
        if (law == LSW) { // LSW: |V|*dt -> accumulated slip
            GatherFaceState(f, dSlipMag);
            Apply(R, out, dSlipMag); // out = R * dSlipMag (separate buffer)
            ScatterAddAccumSlip(f, out);
        } else { // rate-state: state-variable increment
            GatherFaceState(f, dPsi);
            Apply(R, out, dPsi);
            ScatterAddState(f, out);
        }
    }
}
}

```

Phase 4 – Wire into the spatial driver + matrix path

- Expose both knobs in spatial_dyn_driver.cpp; confirm they are honoured on the interior_flux="matrix" (BimaterialWaveOperator) path — these touch only the fault-flux assembly/iterator, not the interior Riemann, so no conflict with the matrix path or with ADER.
- Resample is **on** for TPV31/TPV102/104/205 (symmetric material). Add a guard that **disables resample on any genuine across-fault material contrast** (matching SeisSol's BiMaterial-Fault, Section 4.2).

Acceptance criteria

- Both knobs off $\Rightarrow$ byte-exact vs current (regression contract preserved). `--fault-resample` alone (no over-integration) $\Rightarrow$ also byte-exact, because $R = I$ (Phase 2 test iv).
- On (Phase 0 harness, p1, serial, `--fault-overint on`, then `--fault-overint --fault-resample`): $\max_F |\sigma_n - \sigma_{n0}|$ drops by $\geq 10\times$ and $\max_F \|\llbracket v_n \rrbracket\| \rightarrow$ round-off, with over-integration carrying the per-step reduction and resample suppressing the residual secular creep.
- On (p1 vs p2, both knobs on): the residual σ_n error now **decreases** with order.

7. Validation plan

1. **Local A/B (Phase 0 harness):** off vs `--fault-overint` vs `--fault-overint --fault-resample`; report $\sigma_n / \|\llbracket v_n \rrbracket\|$ curves and the p1 $\rightarrow$ p2 convergence.

```
# Phase 0 local A/B – serial, planar fault, no production run (compile + unit test)
make seas_test_fault_planar_serial
./seas_test_fault_planar_serial --order 1 # baseline: sigma_
./seas_test_fault_planar_serial --order 1 --fault-overint 2 # per-step speckle
./seas_test_fault_planar_serial --order 1 --fault-overint 2 --fault-resample # + secular state
./seas_test_fault_planar_serial --order 2 --fault-overint 2 --fault-resample # order now helps
# NB: --fault-resample ALONE (no over-integration) is a NO-OP: R = identity when
#      #GP == #DOF (Section 4.2), so it must be combined with --fault-overint.
```

2. **Frontera confirm (cheap, flex):** TPV102 pure-upwind ADER with the knobs on — predict σ_n flattens toward 120 MPa like the mixed-flux+RK45 control (job 7760376) but on the **pure-upwind ADER** path.
3. **TPV31:** the decisive target — pure-upwind + ADER + matrix with the knobs on; predict constant σ_n , dip slip $\rightarrow 0$, completion to 15 s.

8. Risks, uncertainties, and fallback

- **Sufficiency is not guaranteed.** Over-integration + resample remove the *source* of the speckle (the per-step flux aliasing and the secular out-of-space state accumulation, Section 3.3). If a *second* path also feeds $\llbracket v_n \rrbracket$ — e.g. the bulk upwind dissipation on the fault-adjacent faces itself, the mechanism Zhang (2023) cures with mixed flux — then dealiasing alone may only *reduce*, not eliminate, the drift. The Phase 0 A/B is designed to detect this.
- **Fallback (still pure-upwind-compatible-ish):** a **Rusanov / Lax–Friedrichs near-fault flux** on the fault-adjacent faces of the matrix operator — central flux plus a *scalar* dissipation $\frac{1}{2}\lambda_{\max}\llbracket Q \rrbracket$. This is what SeisSol added as `fluxNearFault` (Oct 2024, `CellLocalMatrices.cpp:258`); unlike pure central it stays dissipative, so it is **ADER-stable** (no RK needed) and implementable on the matrix path. It changes the near-fault flux (a mild departure from strict pure upwind) but avoids the imaginary-axis instability of pure central + ADER.
- **Pure central + ADER is excluded:** non-dissipative $\Rightarrow$ imaginary-axis eigenvalues $\Rightarrow$ ADER amplifies them (the documented mixed-flux+ADER runaway). Pure central would require porting RK to the matrix path (two pieces of new C++); de-prioritised in favour of the dealiasing route.

9. References

- **Zhang, Liu & Chen (2023)**, *A Mixed-Flux-Based Nodal DG Method for 3-D Dynamic Rupture Modeling*, JGR Solid Earth 128, e2022JB025817. (SSOs from upwind on asymmetric fault-adjacent meshes; nodal-DG mixed-flux cure.)
- **Pelties, Gabriel & Ampuero (2014)**, *Verification of an ADER-DG method for complex dynamic rupture problems*, GMD 7, 847–866. ($((N + 2)^2$ GP fault over-integration; selective upwind dissipation.)

- **Uphoff (2020)**, PhD thesis, TU München (fault Godunov state Eq. 4.51, imposed-state time integral Eq. 4.60 — referenced directly in SeisSol source).
- **Wollherr, Gabriel & Uphoff (2018)**, *Off-fault plasticity ... modal DG*, GJI 214, 1556–1584. (L^2 projection of fault state back to the polynomial space.)
- SeisSol source (read 2026-06-02): `FrictionSolverCommon.h:182,329`; `CpuImpl/LinearSlipWeakening.h:2` (LSW resamples the slip-rate magnitude; `BiMaterialFault` disables resample); `CpuImpl/RateAndState.h:89` and `CpuImpl/FastVelocityWeakeningLaw.h:144` (rate-state resamples the state-variable increment $\Delta\psi$, not the slip rate); `Solver/FreeSurfaceIntegrator.h:42` (`PolyDegree = ConvergenceOrder - 1`, i.e. polynomial degree $N = \text{ConvergenceOrder} - 1$ — so “degree $\leq \text{ConvergenceOrder} - 1$ ” = degree $\leq N$, the full DOF space); `codegen/matrices/dr_stroud_matrices_{2..7}` (resample matrices, rank $(N + 1)(N + 2)/2$ verified); `Kernels/LinearCK/Neighbor.cpp:71`; `Initializer/CellLocalMatrices.cpp:258`.
- SEAS-MFEM: `dynamic/fault_face_flux.cpp:62`; `dynamic/friction_substep_iterator.hpp:201`; `drivers/spatial_dyn_driver.cpp:217,811`; isolation job 7762948 (pure-upwind+RK45).